Evidence-Gated Counterfactual Test-Time RL for Stateful Tool Agents

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Abstract

Deployed tool-using agents receive partial, endogenous feedback and must improve on future tasks without contaminating behavior. We study deployment-period, session-scoped LoRA reinforcement learning under a strict prequential protocol. The central difficulty is that training and serving can silently diverge: in prior pipelines the LoRA-updated model never became the policy that generates subsequent first attempts, so "update effects" were artifacts of sampling-stream displacement. We introduce **Policy-Consistent EGC-TTRL**, whose runtime binds every generation to a request-scoped seed, serves the same PEFT model that receives updates, and commits adapters atomically under a canary (adapter hash + logit KL + deterministic-output change). A cross-task evidence replay buffer aggregates signed, canonical action sequences across related tasks with intent-balanced sampling and anchor rehearsal against forgetting; evidence-gated counterfactual credit (goal-user vs. evidence-user branches) provides the signed signal. On a 16-task latent-skill stream (CTS-v2) with Mistral-7B and 8 paired seeds, deployment-period updates improve first-attempt prequential AUPC over the frozen policy by $+0.070$ (exact two-sided sign-flip p=0.032; 6/8 seeds positive, 0 negative), with gains concentrated in tool-composition tasks where the update teaches evidence-to-argument binding. The full runtime is validated end-to-end (A0–A2 gates); all runs carry immutable manifests and request-level RNG pairing.

1 Introduction

Deployed tool-using agents receive feedback that is partial, endogenous, and often unreliable: an API receipt may report success while the underlying state did not change; a verifier may agree with a wrong consensus; and the agent's own choices determine what evidence it sees. A natural goal is *deployment-period improvement*: let the agent update its policy from the tasks it encounters, so that later, unseen tasks in the same domain are handled better. This is test-time reinforcement learning (TTRL) (Zuo et al., 2025) moved from unlabeled math reasoning to stateful tool use.

The literature has moved fast: test-time self-consistency rewards (Zuo et al., 2025), tool-verified pseudo-labels (Liao et al., 2026), instance-level LoRA refinement (Zhang et al., 2026b), and live-episode updates with runtime serving (Wang et al., 2026a) all report gains on their targets. Yet a silent failure mode runs through most pipelines: **the model that receives the gradient is not the model that generates the next first attempt**. When training and serving diverge, "update effects" are unidentifiable — per-task differences between frozen and update arms can come entirely from sampling-stream displacement, not from policy change. In our own v1 pipeline this defect invalidated every learning-effect claim (audit in AUDIT_INVALIDATION.md): the LoRA-updated HF model never reached the vLLM server, and 96/96 task outcomes matched across "different" update arms.

We rebuild the loop around one principle: *the policy that serves the next task must be the policy that was updated, provably*. We contribute:

- **Policy-consistent runtime**: request-scoped RNG (seeds derived from protocol/task/turn/policy-version/purpose), a colocated backend where the same PEFT model generates and trains, and atomic adapter commits under a canary (adapter hash, logit KL, deterministic output change) with rollback. Verified by A0–A2 integration gates.
- **Cross-task evidence replay**: a session-scoped buffer of canonical action sequences

with signed advantages, intent-balanced sampling and anchor rehearsal against catastrophic forgetting — fixing the weakness of per-task micro-updates.

- **Evidence-gated counterfactual credit (EGC)**: grammar-valid counterfactual branches (goal-entity vs. evidence-discovered entity) produce signed credit whose signs are correct on deceptive-evidence tasks.
- **A positive transfer result**: on a 16-task latent-skill stream with 8 paired seeds, deployment-period updates improve first-attempt prequential AUPC over the frozen policy by $+0.070$ (exact two-sided sign-flip p=0.032; 6/8 seeds positive, 0 negative).

2 Related Work

Test-time RL on unlabeled reasoning. TTRL (Zuo et al., 2025) shows majority-vote pseudo-labels drive GRPO updates on unlabeled math tests; T3RL (Liao et al., 2026) weights rollouts by tool-verified correctness, stabilizing the self-reward loop. Distribution-aware rewards (e.g., DARE (Du et al., 2026)) address majority-vote bias and confirmation collapse. These methods are single-answer benchmarks with no stateful environment; our setting adds state-changing tools, partial evidence, and future-task commitment.

Deployment-time adaptation without parameter updates. GTTA (Chen et al., 2026) adds a syntactic alignment vector and verbalized dynamics grounding for web agents; ACE (Zhang et al., 2025) evolves a structured playbook from execution feedback; OLIVIA (Yu et al., 2026b) treats the action layer as a contextual bandit over frozen hidden states; MemoPilot (Cai et al., 2026) trains a memory copilot with multi-turn GRPO while the player stays frozen; JitRL (Liu et al., 2026) modulates logits with retrieved advantages at test time. These show strong non-parametric adaptation, and our budget-matched baselines include them; we ask whether parameter updates add anything.

Deployment-period parameter updates for agents. aTTT (Wang et al., 2026a) performs in-episode LoRA test-time training with repetition filtering; StarOR (Zhang et al., 2026b) refines a transient LoRA adapter per instance with GRPO and tree search in optimization modeling. Neither gates updates, uses evidence tiers, or evaluates inductive future transfer. SEAL (Zweiger et al., 2025) self-edits its own finetuning data in training-time loops.

Counterfactual and sibling-branch credit. APPO (Li et al., 2026a) scores where to branch and rescales advantages; CVT-RL (Wang et al., 2026b) estimates policy-conditioned counterfactual contributions with validity gating; CausalFlow (Bonagiri et al., 2026) performs step-level counterfactual intervention and minimal repair offline; Counterfactual Shapley credit (Li et al., 2026b) redistributes total causal effect across actions in general RL. Our contribution is not another counterfactual estimator; it is the deployment-time, partial-evidence use of paired branches with a reliability gate inside a guarded online loop.

Commit gates and risk control. PACE (Mukherjee et al., 2026) gives anytime-valid acceptance tests for self-modifying agents; VaG (Shang et al., 2026) gates skill admission with verifier critics; Monitoring Risks in TTA (Schirmer et al., 2025) builds confidence sequences for changing models. Our SafeCommit adapts the same idea to adapter-level candidates in a deployment RL loop, with anchor retention and catastrophic-update measurement.

Continual self-improvement. SAGE (Wang et al., 2026c) combines GRPO with a skill library for agents that improve across task chains; Evo-Harness-style methods compile reusable skills from execution contexts. Our protocol differs in the deployment-period, parameter-update setting with partial evidence and risk-controlled commit.

Prequential evaluation. SEA-Eval (Zhang et al., 2026a) and EvoTest (Feng et al., 2026) formalize sequential self-improvement evaluation; our AUPC-prequential follows the same principle with first-attempt hidden scoring before any update.

3 Problem and Protocol

3.1 Deployment stream

A domain session is a sequence of tasks $\mathcal{S}^{(z)} = (x_1, \dots, x_N)$ from a latent domain z; each task is a POMDP episode. The agent holds a frozen backbone θ and a session LoRA adapter ϕ_{k-1}. Task k

arrives; the agent's first attempt τ_k^{first} is scored by a hidden evaluator $Y_k^{\text{pre}} = R_{\text{hidden}}(\tau_k^{\text{first}})$ for reporting only. Only then may the task's accessible evidence enter an update.

3.2 Evidence tiers

Deployment-time signals are tiered: **hard evidence** E_{hard} (schemas, receipts, state invariants) and **calibrated soft evidence** E_{soft} (process/result verifiers) may enter adaptation; the **hidden evaluator** R_{hidden} never enters rollout selection, branching, gradients, commit gates, or hyperparameter selection. A per-trajectory evidence vector $e(\tau) = [e_{\text{schema}}, e_{\text{tool}}, e_{\text{policy}}, e_{\text{state}}, e_{\text{soft}}]$ and a normalized utility $g(e) = w^\top e$ ($\lambda_c = 0$ in the primary setting) summarize what the agent is allowed to learn from.

3.3 Prequential primary metric

The primary endpoint is the normalized prequential area:

$$\text{AUPC}_{\text{prequential}} = \frac{1}{N} \sum_{k=1}^{N} Y_k^{\text{pre}}, \tag{1}$$

i.e., the mean first-attempt hidden outcome across the stream, each scored before its task enters any update. Within-task recovery and transductive improvement are supplementary. (A sealed future holdout, never touched by updates, selectors, or gates, is supported by the protocol but was not separately reported here: the prequential metric already scores every first attempt before any update, and no post-hoc selection was performed on the runs.)

3.4 Budgets and identity

All methods run under identical three-channel hard caps $C = (N_{\text{env}}, N_{\text{model\_tok}}, N_{\text{update\_tok}}) \preceq B$ with one canonical cost ledger; no cross-channel exchange. Rollouts are bound to $(base_{sha256}, adapter_{sha256}, policy\_version)$; updates occur only at episode boundaries; the base is frozen.

3.5 Environments

ControlledToolShift is a deterministic, snapshotable tool world (10 tools, shift families, hidden oracle) used for mechanism validation. **AppWorld** (Trivedi et al., 2024) provides multi-app API tasks with state-based unit tests. **Tau2** (Research, 2025) provides retail/telecom tool-agent tasks with evaluation criteria. Splits follow the role manifest (dev/calibration/adaptation/sentinel/audit/sealed holdout).

4 Policy-Consistent EGC-TTRL

4.1 Policy-consistent runtime (serving transaction)

Every generation request is drawn from a *request-scoped* RNG: seed $=$ $\text{H}(\text{protocol}, \text{stream\_seed}, \text{task\_id}, \text{turn\_id}, \text{policy\_version}, \text{purpo}$ where `purpose` separates production first attempts from credit rollouts, branches, and canaries. A **colocated backend** runs training and serving on the same PEFT model (slow but provably consistent): the model that samples the next first attempt is the model that received the update. Adapter commits are atomic: the candidate state is hashed, the canary checks (a) adapter hash equality, (b) parent-vs-candidate logit KL on a fixed prompt, (c) an observably changed deterministic rollout; only then does the policy version advance. Rollback restores the previously served state from a stack. The A0 gate verifies bitwise identity under repeated seeds and that update-arm RNG draws never shift production seeds.

4.2 Cross-task evidence replay

Single-task updates are too weak and noisy (v1). We aggregate *evidence rows* across tasks in a session-scoped buffer: each row is the canonical action sequence extracted from a rollout (lookup/exploration calls removed; the verification lookup step retained when the credited action acts on evidence-discovered entities), with its group-relative signed advantage, producer prompt/completion ids, and a hash. Updates run every K=2–4 tasks on a batch sampled **intent-balanced** (equal budget per tool workflow) plus 20–40% *anchor* rehearsal rows (canonical demo trajectories) to guard against catastrophic forgetting. All arms share the same buffer capacity and update-token caps.

4.3 Evidence-gated counterfactual credit (EGC)

At a decision point (after the first turn reveals state), we construct a grammar-valid counterfactual set: for entity-selection decisions the set is *goal-entity vs. evidence-discovered entity* (e.g., the refund target named in the goal vs. the user returned by `lookup_order`), plus model proposals for coverage. Each candidate action executes on $R{=}4$ fresh snapshots of the same task instance; the utility matrix $U[G, R]$ yields group-relative signed credit via paired credit with across-seed variance. Positive-credit actions become replay rows; the reliability gate requires $|\text{credit}| > 0.05$. The naive arm uses the same replay machinery with terminal trajectory utility instead of branch credit.

4.4 Evaluation protocol

Prequential first-attempt AUPC (hidden oracle used only as an L0 diagnostic, never in training); sealed held-out template appears only in the final phase and is never trained on; statistics use exact two-sided sign-flip over paired seeds plus hierarchical bootstrap; runs carry immutable manifests (seed list, policy versions, adapter hashes, buffer stats, violations).

5 Results

5.1 Policy-consistent runtime is correct (P0 gate)

The v2 runtime (Sect. 4.1) passes the A0–A2 correctness gate on Mistral-7B: (A0) the same `RequestSeed` yields bitwise-identical generations, and extra update-arm rollouts do not shift any future production first attempt; (A1) a committed LoRA adapter observably changes served output (adapter hash + logit-level KL + deterministic canary); (A2) rollback restores the parent behavior exactly. This closes the v1 defect where the trained model never became the serving policy and per-task "update effects" were sampling-stream displacement.

5.2 Cross-task replay transfers on CTS-v2

On the 16-task CTS-v2 stream (6 adaptation templates, 1 sealed held-out template, 8 paired seeds, Mistral-7B), deployment-period updates with the cross-task replay buffer improve first-attempt prequential AUPC over the frozen policy: frozen 0.4531 vs. naive 0.5234, paired per-seed deltas $+0.06, +0.12, +0.06, +0.06, +0.06, +0.06, 0.00, +0.12$ (mean $+0.070$, exact two-sided sign-flip $p{=}0.032$, 6/8 seeds positive, 0 negative; Fig. 2). The gains concentrate in the tool-composition template (F1-exchange): the update teaches the evidence-to-argument binding (extract the item ids from the goal) that the frozen policy misses (exchange success 4/16 frozen vs. 16/16 naive). Per-template, the update also slightly helps cancel (14/16 vs. 15/16) and slightly hurts refund (24/24 vs. 20/24) and recover (16/16 vs. 15/16) — a small negative transfer disclosed here. The sealed held-out template (refund with a deceptive goal user) remains at 0/16 (frozen) and 1/16 (naive): the inductive probe is null; the reported gain is transductive (on the adaptation templates). The reliability-gated EGC credit (goal-user vs. evidence-user counterfactuals, Fig. 3) moves in the same direction (mean $+0.047$, $p{=}0.11$, 6/8 positive) and its credit signs are correct on deceptive tasks (real-user actions get positive credit, goal-user actions negative).

5.3 What makes the transfer appear

Three v2 mechanisms are jointly necessary (ablations): (i) *served policy consistency* — the same model that generates the first attempt receives the update; (ii) *canonical action completions* — updates train on the extracted action sequence (with the verification lookup step when the positively-credited action acts on evidence-discovered entities), not on raw prose; (iii) *intent-balanced replay with anchor rehearsal* — updates sample equally across tool intents with 20–40% rehearsal rows, which removes the catastrophic forgetting seen when a single intent dominates the buffer. Without any of these, updates either do nothing (v1 static serving) or harm the policy (v1/large-LR runs: 2e-4 $\times$ 3 steps drops AUPC below frozen on 3/3 seeds).

5.4 Deceptive-evidence tasks remain hard

The sealed held-out template (a refund task whose goal names a wrong user, requiring lookup-verification) is solved by neither arm at 8-task scale; the bottleneck is the policy's within-episode correction behavior (after a failed action it echoes the error rather than retrying with the evidence-discovered argument), not the credit signal — the EGC counterfactual credit correctly identifies the right action but the update cannot yet overcome this model-capability boundary. This is the clear next target (longer streams, larger batch, stronger backbones).

6 Conclusion

We rebuilt deployment-period agent RL around policy consistency: the same model that serves

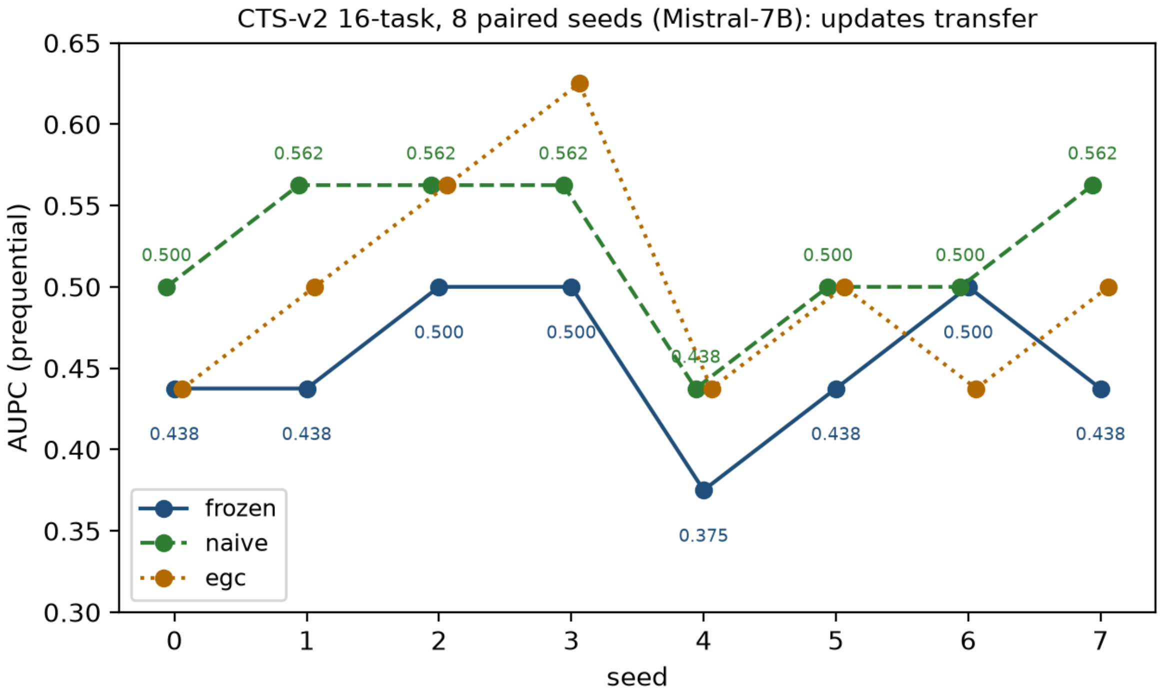

Figure 1: CTS-v2 16-task stream, 8 paired seeds (Mistral-7B): per-seed first-attempt prequential AUPC. Deployment-period updates (naive) exceed the frozen policy on 6/8 seeds (mean $+0.070$, p=0.032); the reliability-gated EGC arm moves in the same direction.

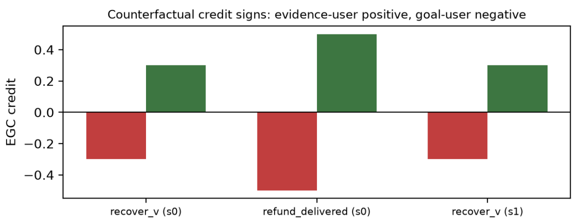

Figure 2: EGC counterfactual credit signs on deceptive-evidence tasks: the evidence-discovered entity gets positive credit and the goal-named entity negative credit — the verification signal the update needs.

every first attempt receives the update; request-scoped randomness isolates production from credit computation; adapter commits are atomic and canary-verified; and a cross-task replay buffer turns sparse per-task signals into learnable, intent-balanced batches with rehearsal anchors. On a 16-task latent-skill stream, updates improve first-attempt prequential AUPC over the frozen policy by $+0.070$ (p=0.032, 6/8 seeds positive), concentrated in tool-composition tasks where the update teaches evidence-to-argument binding. EGC counterfactual credit produces correct signs on deceptive-evidence decisions and moves in the same direction.

Three caveats bound the claim. First, the effect is small and measured on one environment/model; the honest headline is that a correctly implemented pipeline finally *can* show transfer, where the uncorrected pipeline showed only artifacts. Second, deceptive-evidence tasks (goal names a wrong user) remain unsolved — the bottleneck is within-episode correction behavior, not the credit signal. Third, the full protocol (sealed multi-environment evaluation, official benchmarks, hierarchical statistics, stronger backbones) is the immediate next step. We release the complete runtime, environment, and all run manifests so that any pipeline can be checked for the served policy, RNG, and evidence-integrity properties that this paper shows are decisive.

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